

NJS_Q06_M: Memory Leak Diagnostics

Difficulty: MEDIUM

Scenario

A Node.js web application is exhibiting runtime anomalies or configuration errors in production. You are tasked with writing clean, async, and correct JavaScript code to solve the requirements.

Username & Naming Conventions

- Resources must utilize your candidate suffix prefix (<username>-<exam_code>).

Task Objectives

- Node.js Optimization & Code Resolution:** Diagnose and patch a memory leak caused by unbounded array accumulation in Express route handlers.
- Ensure your changes pass all local verification tests.

Verification

- Local run test script `run.py` can be executed to verify local workspace properties.
- Central validation runs queries against the live workspace to verify integration.

Grading Criteria

- TC1: File `server.js` exists in workspace** (4 Marks)
- TC2: Memory leak diagnostic report created** (4 Marks)
- TC3: Leaking array resolved or replaced with size-limited collection** (4 Marks)
- TC4: Express routes perform successfully under load** (4 Marks)
- TC5: Memory heap consumption verified to remain bounded** (4 Marks)

Important Notes

- Always check that your syntax compiles without errors.